

1.2.1 Where It Fits In

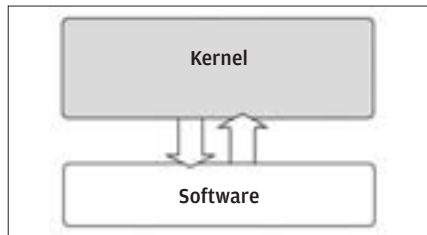
The operating system's kernel sits in the system memory in a protected space all to itself, called the *Kernel Mode*. It is this memory space that is used for the many functions it performs. The remaining space is called the *user mode*, and this is used by the various applications that you run on your OS. None of these applications are allowed in the kernel's private space; this prevents them from corrupting the kernel's data and crashing the system.

When your system starts up, the kernel is loaded into the memory, and remains there till the system shuts down.

Designing a good, robust kernel has always been a challenge, and OS developers use different approaches to get what they want.

1.2.2 The Lone Ranger—The Monolithic Kernel

The monolithic kernel is the one-stop solution for all the features of the OS. All these features run in the kernel mode. The features are written in *modules*, which are logical divisions of the different functions of the OS. So your OS would have a Memory Management Module, a Disk Management Module, and so on. Software talks directly to the kernel, which talks directly to the hardware. The monolithic approach is quick and efficient because of the tight integration between all its different modules; it makes optimum use of the resources at its disposal.



The Monolithic Kernel - the simplest way for software to talk to the hardware

However, there are dangers to this approach. It is very difficult to write kernel modules that gel well with each other, and even then, the tiniest mistake in even the most insignificant module is capable of taking the whole system down. Another problem with the monolithic kernel is that it can be quite a waste, because it